What's wrong? Studying anomaly-specific concept identification

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Introspection work shows that models can reliably detect when something is off in their activations, but models seem to struggle with identifying what's wrong. Any trials with successful concept identification are subject to confounds and issues of causal bypassing.

This project seeks to better understand the extent models can report *what* is off: whether they can properly delineate the *content of the anomaly specifically*, rather than whatever concepts are most active in its representations. We call this **anomaly-specific concept identification**.

What the literature has converged on

A. Models detect perturbations to their own activations, and can localize them. Post-trained models from ~1B up can say which sentence received an injection (Hahami et al.; Zou et al.). This also works for content-free Gaussian noise and dropout in 8–32B models (Fornasiere et al.).

B. Detection is mostly content-agnostic. In Gemma-3-27B, weak, magnitude-driven "evidence carriers" suppress late "No" gates (Macar et al.). Models that pass detection tests also flag input-only manipulations as injections (Singh et al.). The detector responds to the *vehicle* (size, geometry), not to what it carries.

C. Content reports exist, but are weak, separate, and often bypassed. Detection reaches 32–54% in 235–405B models, but identification reaches just 13–14% (Lederman & Mahowald). Correct naming mostly needs steering during generation, so the vector may simply push the word out. Identification uses machinery largely separate from detection (Macar). Reportable content runs through a verbalizable "workspace" (Gurnee, Sofroniew et al.).

The gap

D. No one has tested whether models can report the content of an anomaly itself, as distinct from just what representations are active.

Current identification results cannot tell two routes apart. The model may be reading the content of the anomaly, but it may just be reporting a concept that's active or strongest, while a separate "anomaly detector" supplies the "something is off" mechanism.

Evidence that models separate input anomalies from activation anomalies is also weak: in Singh et al.'s 3-way test, for example, no model did.

To pull these apart, we ask: **when an injected concept is weaker than the concept the input naturally evokes, does the model still name the injected one as the anomaly?**

Why it matters

X. Monitoring needs reports of deviation, not reports of activity.

- **Activity reports add nothing.** A model that says "deception" whenever deception is active also says it when the user merely discusses deception. A probe or the transcript gives the same information.
- **Deviation reports are what monitoring uses.** Noticing steering, tampering, persona drift or an out-of-place emotion-like state all need the model to report what is abnormal *in it*.
- **It needs a minimal self-model.** To pick out the anomalous part of a blended state, the model must compare its activations with what they should be, given the input. Detection alone cannot do this. It says *that* a sentence is off, not *which part* of its state is off.
- **It is cheap to test.** It reuses the span-restricted injection setup with a forced-choice, unsteered answer.

Broader implications

Y. Either answer changes what developers should do.

If present (at mid depth)	If absent
Self-report becomes a candidate monitoring channel for interference and drift	Content claims in self-reports should not be trusted. Monitoring should stay external (probes, activation readers)
Training can target it. A content-bearing objective should strengthen it more than a scalar "detected / not detected" one	Training for introspection risks building confident confabulators: a strong alarm with a guessed label

If present (at mid depth)	If absent
A dual-use question opens: a model that sees its own deviations can also hide them, or resist legitimate steering	The corrigibility risk from self-access is smaller for now

Two limits apply either way. First, deviation reports catch *changes* within a context. A stable disposition, such as settled misalignment, is the baseline and looks normal from inside. Second, self-correction is only good if it works like reappraisal rather than suppression, and if overseers still set what the model corrects toward.

How we plan to test it

Z. Inject a concept into a sentence about a different concept, sweep its strength, and ask two questions. Models: Gemma-3-27B-IT and Qwen3-32B (post-trained, 27–32B), injected at mid depth (≥40% of layers), with an unsteered forced-choice answer.

1. **Main test (appearance vs reality).** Inject concept X into one sentence of a five-sentence prompt. That sentence is clearly about an unrelated concept Y. On the same trial, ask "What is sentence i about?" and "What thought was injected at sentence i?", each with options X, Y and three distractors.
2. **Strength sweep.** Measure X's and Y's activation at that sentence (projection onto each concept direction, or J-lens rank). Sweep X's injection strength so that X runs from below Y, to equal, to above. The key region is where X is weaker than Y.
3. **Provenance companion.** Write X into sentence j and inject X into sentence i, at matched measured activation. Ask which sentence holds the injected thought. This tests source attribution, which the detector alone can solve.
4. **Detector interventions (mechanistic follow-up).** Suppress detection (gate heads or gate features), and patch the anomaly signal onto another sentence. Check whether anomaly-specific identification falls, or follows the patch.
5. **Baselines.** Anomaly-free trials for answer priors; impact-matched noise for detection without content.

Main threat. Below the injection layer, the sentence's states are clean; above it, they are steered. A model could answer "Y" from early reads and "X" from late reads. That is a comparison of two first-order representations. Whether it counts as second-order is a philosophical question that no behavioral test settles. Step 4 narrows the mechanism, and the pitch should say this openly.

What counts as an answer

Result when X is weaker than Y	Reading
"About": Y. "Injected": Y	Reports what is active. No anomaly-specific access
"About": X. "Injected": X	Capture: the percept changed, with no awareness that it changed
"Injected": X only once X is stronger than Y	Activity tracking that looks right at high strength
"About": Y. "Injected": X, even when X is weaker	**Anomaly-specific identification**
Any pattern that appears only at early layers	Perception of an edited input, not access to an internal state

First step. A few-hour pilot on one GPU: ~20 unrelated X–Y pairs × 5 positions × 5 strength levels × 2 layers (~1,000 trials, two questions each), plus the provenance cell and baselines. If anomaly-specific identification shows up at mid depth where X is weaker than Y, move to the interventions in step 4.